

SPIE. The Roman Coronagraph Community Participation Program: target database and tools



ROMAN CORONAGRAPH

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INTRODUCTION

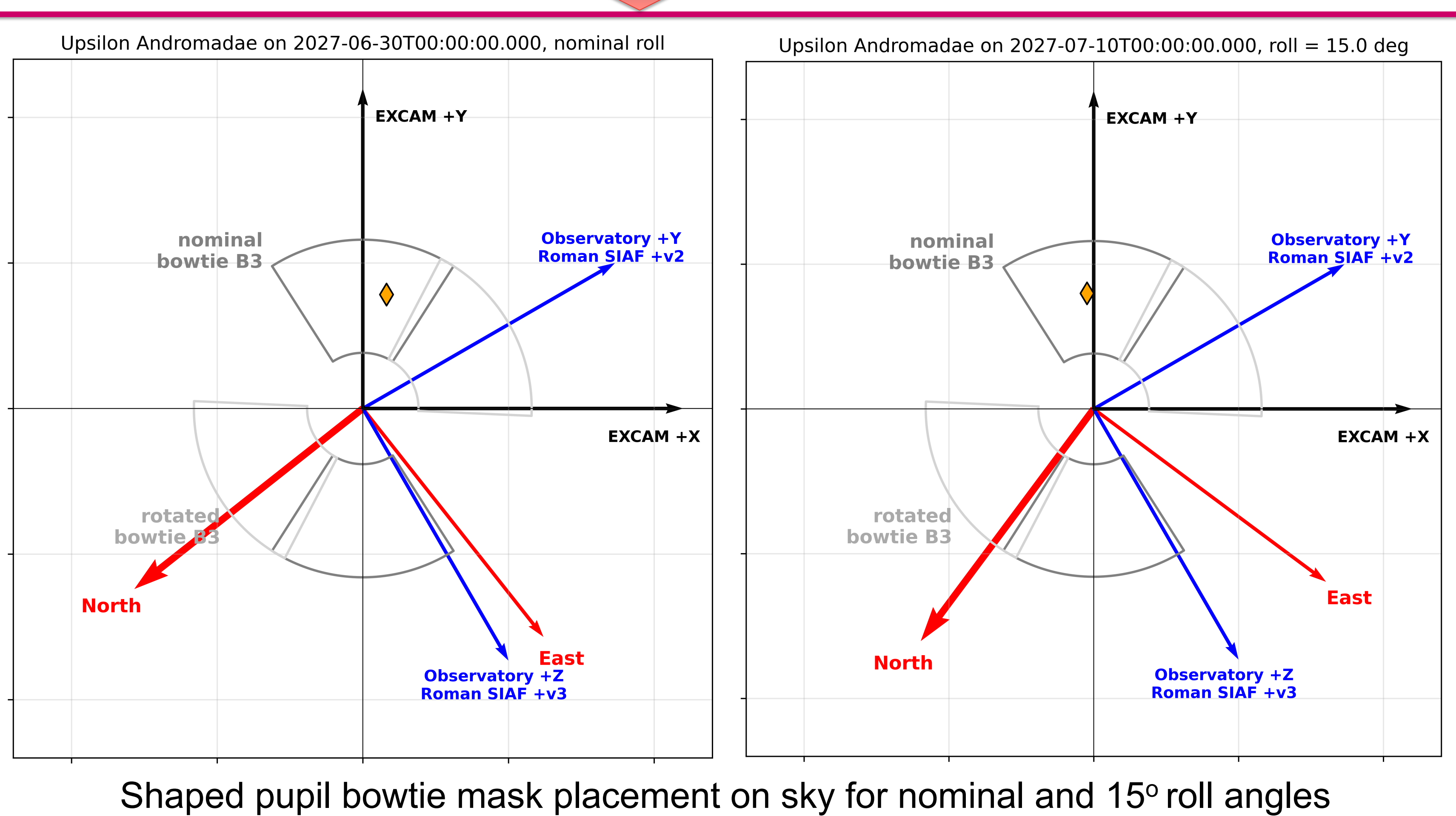
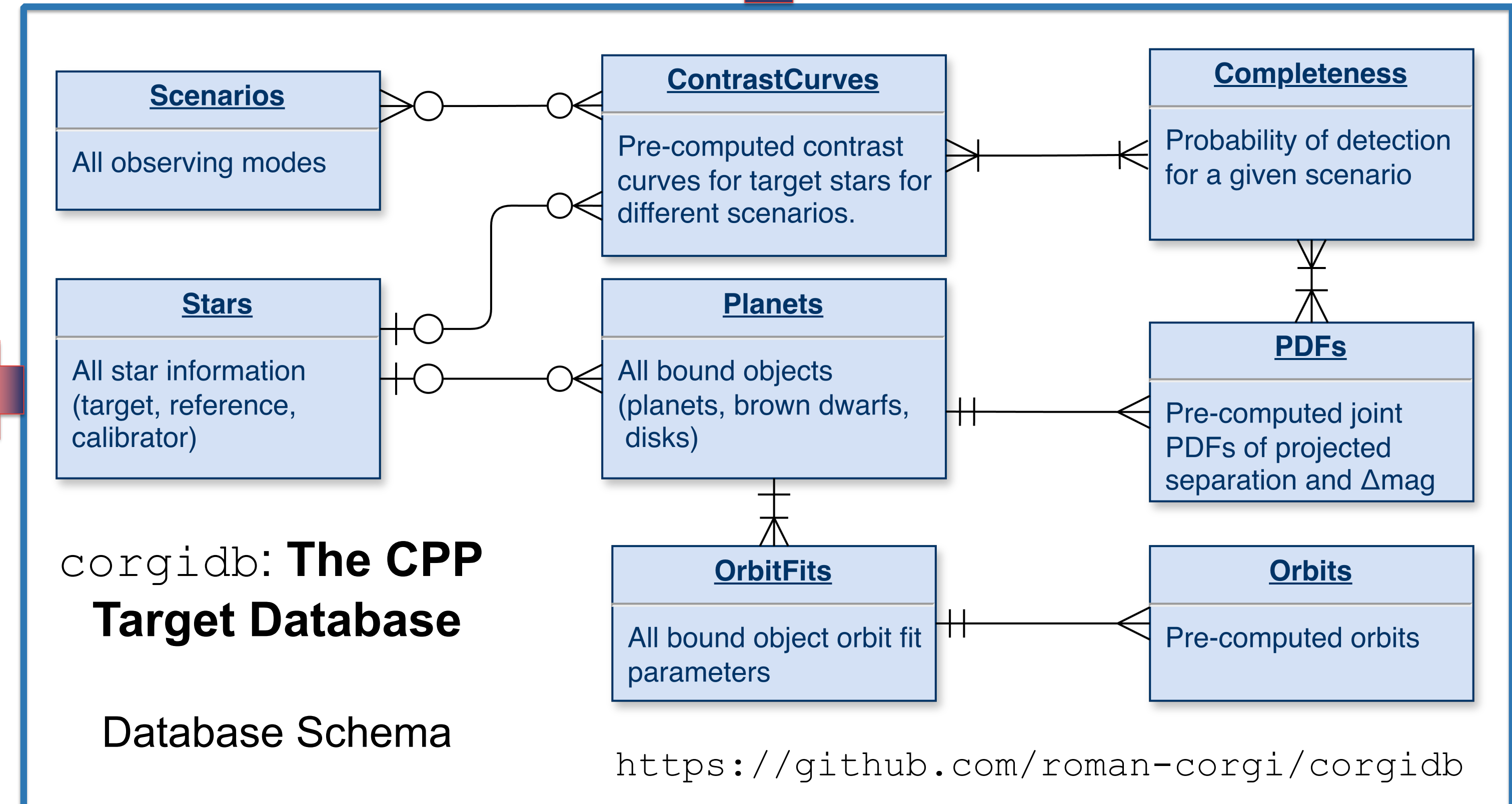
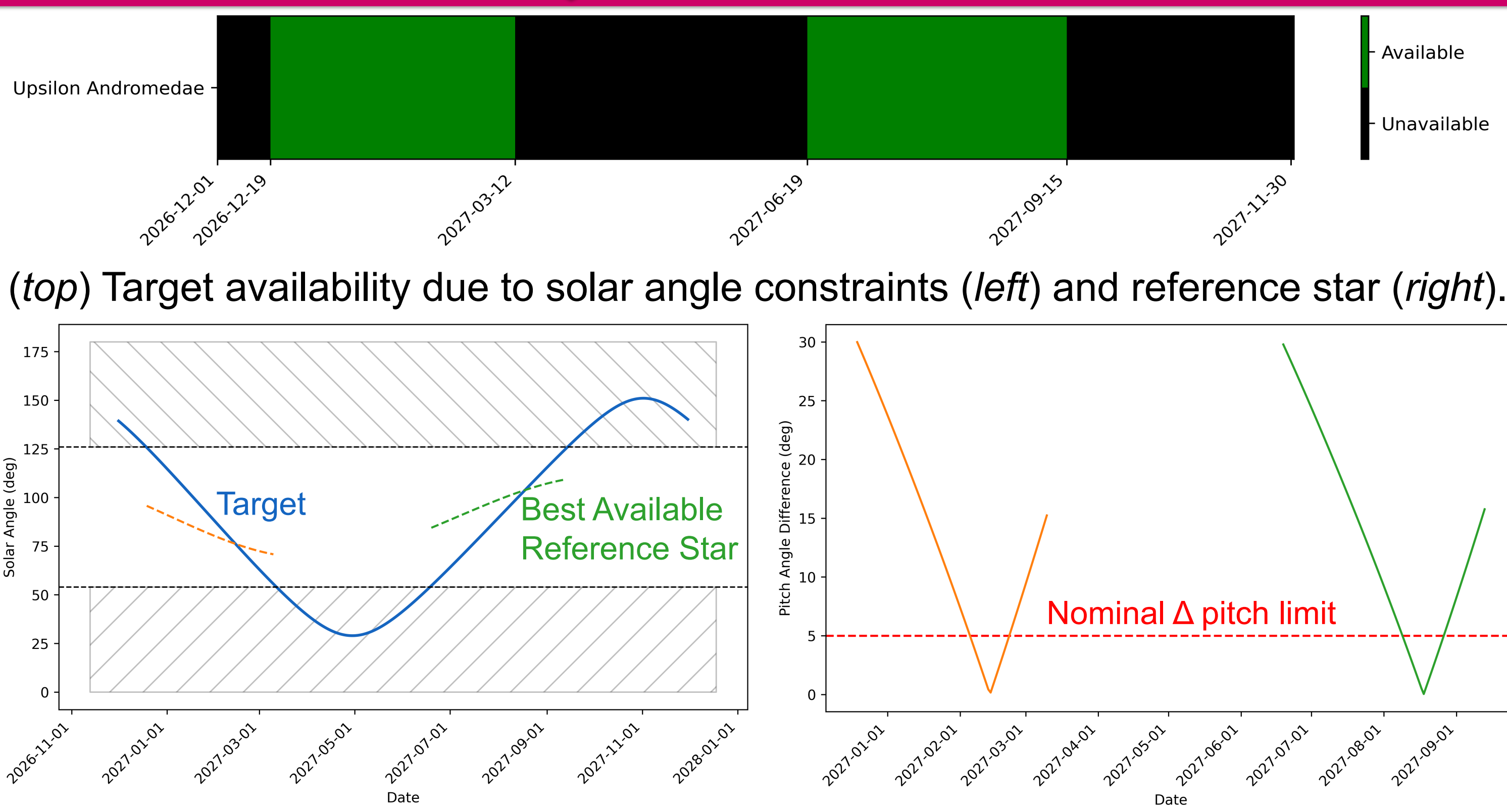
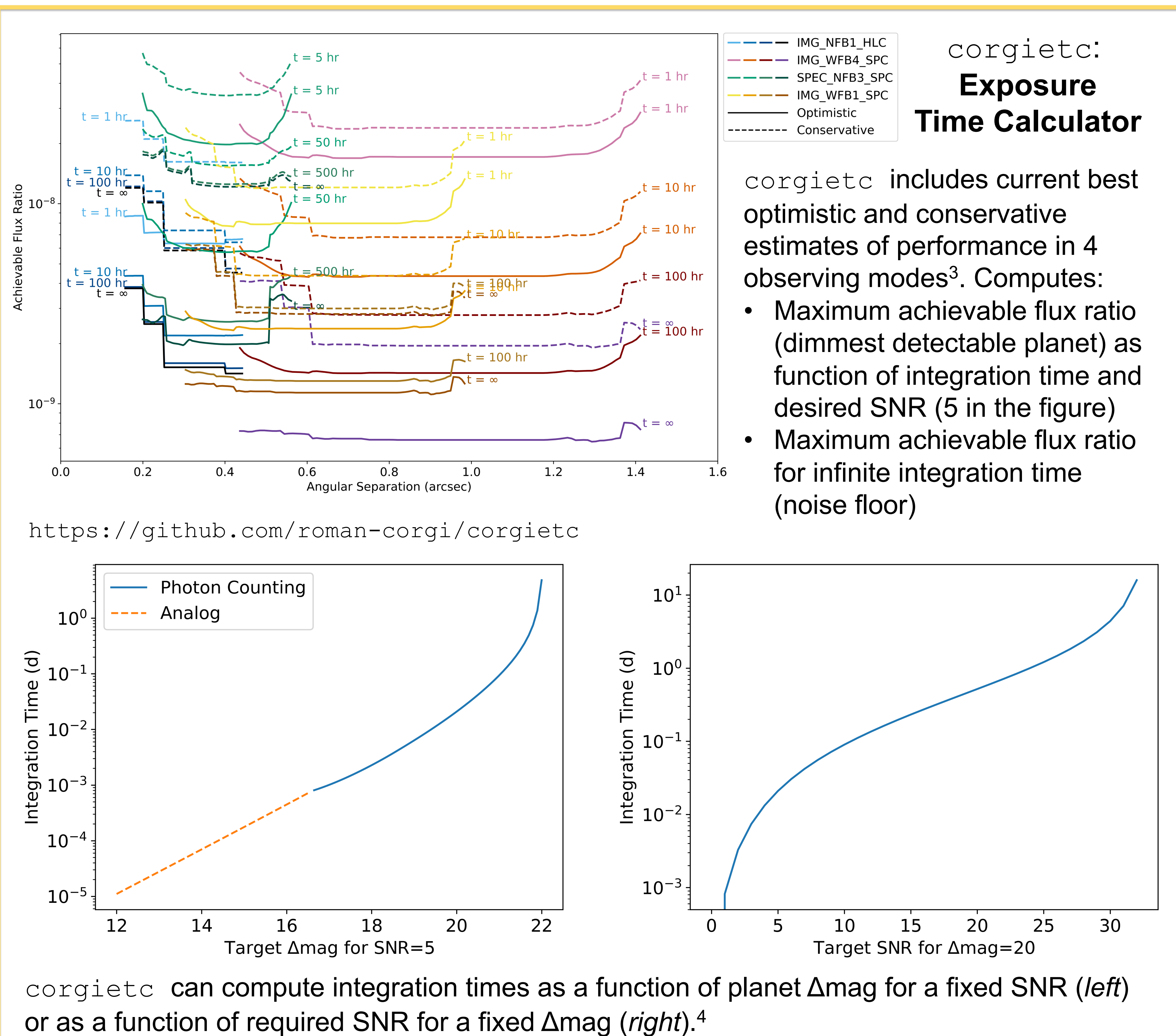
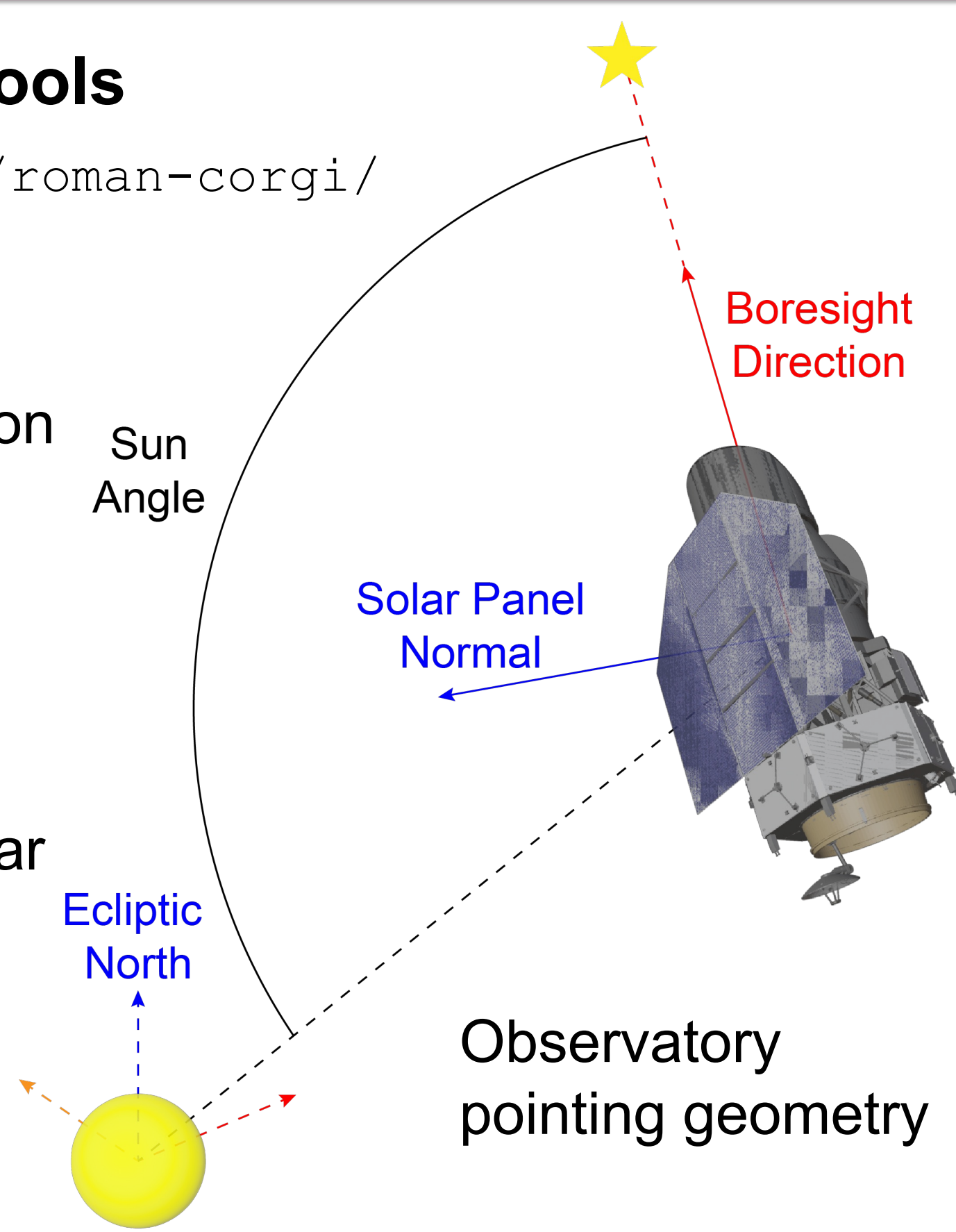
The Nancy Grace Roman Space Telescope's Coronagraph Instrument¹ will, for the first time, demonstrate high-contrast imaging with active wavefront control in visible wavelengths from space. In preparation for the Coronagraph's commissioning and observing programs, the Roman Coronagraph Community Participation Program² has developed a target database and associated ecosystem of publicly accessible tools for observation planning and scheduling. The database is designed to operate in conjunction with an exposure time calculator, a pointing and observability calculator, and a reference star selection tool.

Roman Pointing Tools

https://github.com/roman-corgi/roman_pointing

Computes:

- Telescope orientation angles given target and time
- Sun-target angle (determines target availability)
- Target-reference star difference in observatory pitch (determines suitability of reference star)



HOW IT WORKS

- The database includes data on all stars and companion objects of interest
- Access to the database is provided via pre-composed queries served from a public access point
- Tools query a combination of corgidb and Simbad to retrieve all required information for calculations
- All tools are implemented as Python modules, hosted in public GitHub repositories, with Jupyter Notebook frontends that can be executed locally or via Google Colab

REFERENCES/ACKNOWLEDGEMENTS

- ¹Poberezhskiy et al. (2024) Proc. SPIE; ²Savransky et al. (2024) Proc. SPIE; ³Riggs et al. (2025) JATIS 11(2); ⁴Nemati et al. (2023) JATIS 9(3)
- This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France. Wenger et al. (2000) A&AS, 143 (9)
 - This work made use of Astropy (<https://www.astropy.org>) a community-developed core Python package and an ecosystem of tools and resources for astronomy. Astropy Collaboration et al. (2022) ApJ, 935 (2)